ISOM 3540 – Introduction to Probability

Assignment 3: Solution to Question 5

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Submission deadline: 30 October 2026

Data and modeling conventions. The S&P 500 *daily closing price index* P_t is used for every US market trading session from 4 January 2021 through 31 December 2025 ($N = 1{,}255$ returns). The preceding close is $P_{2020\text{-}12\text{-}31} = 3756.07$. In accordance with the question,

$$R_t = 100\left(\frac{P_t - P_{t-1}}{P_{t-1}}\right), \tag{1}$$

where $t-1$ means the *preceding trading session*. A value $R_t = -1.5$ denotes a -1.5% return; volatility and VaR are in percentage points (pp), and variance is in pp^2. Missing market holidays are omitted, not treated as zero returns. The index is a price index (dividends excluded). Data sources are given at the end; all charts and daily calculations are available in the companion Excel workbook.

Since Question 5 does not specify the RiskMetrics decay factor, initialization, or degrees of freedom for the numerical illustration, the following assumptions are made explicitly: $\lambda = 0.94$ (standard one-day RiskMetrics setting); the initial variance is the mean squared return over the *last 60 trading sessions of 2020* (7 October–31 December 2020); and $\nu = 5$ for the numerical standardized-t VaR comparison. We assume a zero conditional return mean, as in the simple RiskMetrics approach. These choices are held fixed for all 2021–2025 forecasts.

(a) RiskMetrics variance in 2021–2025

At the beginning of session t, the one-day-ahead variance forecast is

$$\boxed{\sigma_t^2 = 0.94\sigma_{t-1}^2 + 0.06R_{t-1}^2}, \qquad \sigma_t = \sqrt{\sigma_t^2}. \tag{2}$$

Only information through $t-1$ enters this forecast, avoiding look-ahead bias. For the first observation, using the previous 60 returns yields $\sigma^2_{2021\text{-}01\text{-}04} = 1.032150\,\text{pp}^2$. For illustration, since $R_{2021\text{-}01\text{-}04} = -1.475478$,

$$\sigma^2_{2021\text{-}01\text{-}05} = 0.94(1.032150) + 0.06(-1.475478)^2 = \boxed{1.100843\,\text{pp}^2}.$$

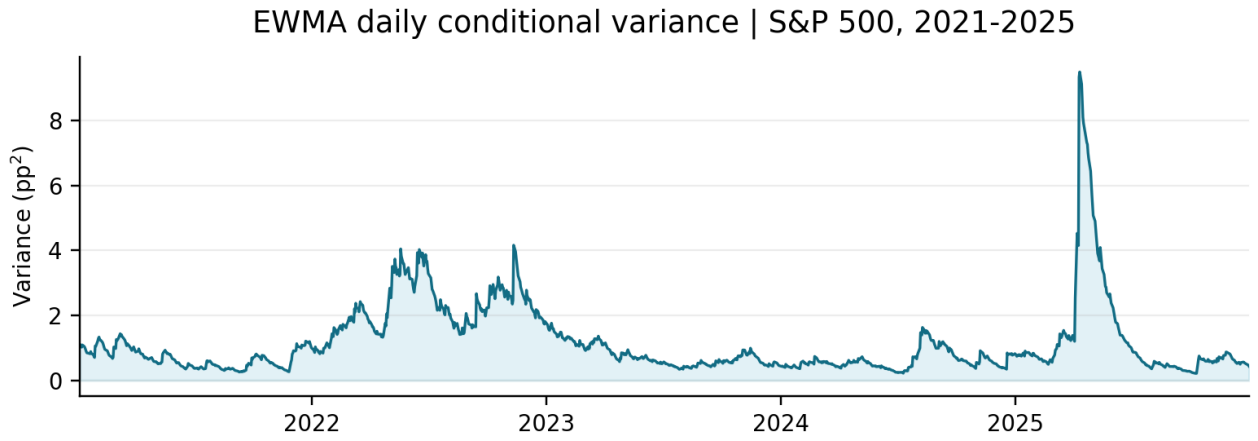


Figure 1: Daily EWMA forecast variance σ_t^2 from 2021 to 2025; replicated as an editable native Excel chart.

(b) Normal distribution: 5% and 1% daily VaR and backtesting

For a long S&P 500 position with conditional mean zero, define $\mathrm{VaR}_{\alpha,t} > 0$ as the percentage *loss magnitude* such that $P(R_t < -\mathrm{VaR}_{\alpha,t}) = \alpha$. If $R_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$, then

$$\boxed{\mathrm{VaR}^N_{\alpha,t} = -\Phi^{-1}(\alpha)\sigma_t}, \quad \begin{cases} \mathrm{VaR}^N_{5\%,t} = 1.644854\,\sigma_t, \\ \mathrm{VaR}^N_{1\%,t} = 2.326348\,\sigma_t. \end{cases} \tag{3}$$

A VaR exception occurs on date t if the *realized* daily return $R_t < -\mathrm{VaR}_{\alpha,t}$. The target number of exceptions over $N = 1255$ days is $N\alpha$: 62.75 for 5%, and 12.55 for 1%.

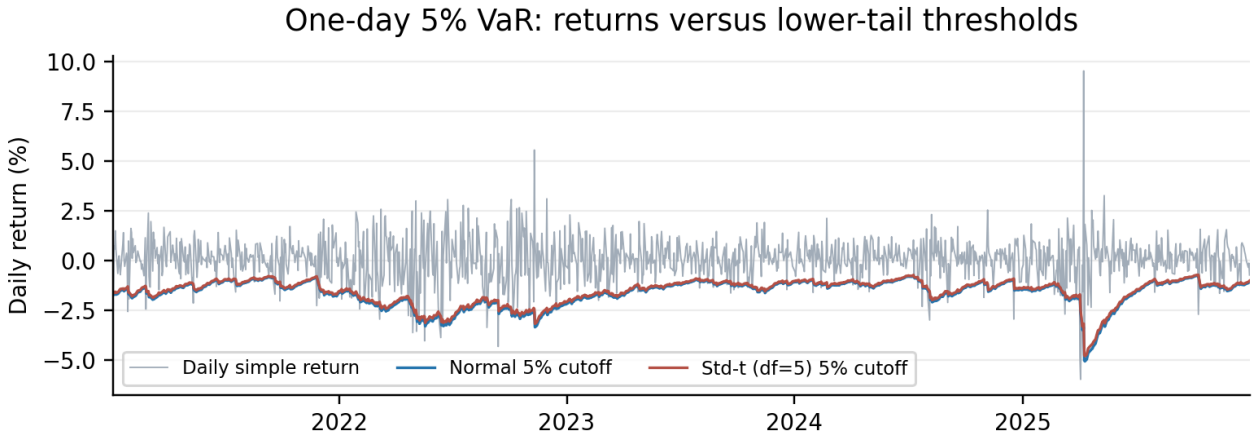


Figure 2: Daily returns and negative 5% VaR thresholds, normal and standardized-t models. A breach occurs when the gray realized-return line is below the relevant threshold.

		Normal model exceptions		Standardized-t exceptions	
Year	Days	5%	1%	5%	1%
2021	252	12	8	12	7
2022	251	20	4	22	2
2023	250	13	2	15	0
2024	252	14	6	16	5
2025	250	15	6	16	5
Total	**1255**	**74**	**26**	**81**	**19**

Performance criterion. To assess coverage, let K denote the total number of exceptions and $\hat{p} = K/N$. Under the null hypothesis $H_0 : p = \alpha$, the Kupiec unconditional-coverage likelihood-ratio statistic is

$$LR_{\mathrm{UC}} = 2\left[K \ln\left(\frac{\hat{p}}{\alpha}\right) + (N-K)\ln\left(\frac{1-\hat{p}}{1-\alpha}\right)\right] \overset{H_0}{\approx} \chi^2_1. \tag{4}$$

For the **normal 5% VaR**, the observed rate is $74/1255 = 5.90\%$, $LR_{\mathrm{UC}} = 2.013$, and $p = 0.156$: *do not reject* correct coverage at the 5% significance level. For the **normal 1% VaR**, the observed rate is $26/1255 = 2.07\%$, $LR_{\mathrm{UC}} = 11.122$, and $p = 0.00085$: *reject* correct coverage; this VaR substantially underestimates extreme downside risk.

(c) Standardized Student's t density and percentiles

Let $X \sim t_\nu$ be an ordinary Student's t random variable with $\nu > 2$ degrees of freedom and density

$$f_X(x) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}, \qquad x \in \mathbb{R}.$$

Define $Y = cX$ where $c = \sqrt{(\nu-2)/\nu}$. Because $x = y/c$ and $dx/dy = 1/c$, the change-of-variables formula gives

$$\begin{aligned} f_Y(y) &= \frac{1}{c} f_X\left(\frac{y}{c}\right) \\ &= \boxed{\frac{\Gamma((\nu+1)/2)}{\sqrt{(\nu-2)\pi}\,\Gamma(\nu/2)} \left(1 + \frac{y^2}{\nu-2}\right)^{-(\nu+1)/2}}, \qquad -\infty < y < \infty. \end{aligned}$$

As $\mathrm{Var}(X) = \nu/(\nu-2)$, it follows that $E(Y) = 0$ and $\mathrm{Var}(Y) = 1$. Let $t_\nu^{-1}(\alpha)$ denote the α-quantile of the ordinary t_ν distribution. Since $c > 0$, the required percentiles of the *standardized t* distribution are

$$\boxed{q_{0.01}(Y) = \sqrt{\frac{\nu-2}{\nu}}\, t_\nu^{-1}(0.01), \qquad q_{0.05}(Y) = \sqrt{\frac{\nu-2}{\nu}}\, t_\nu^{-1}(0.05)}. \tag{5}$$

The expressions above hold for any $\nu > 2$. For the numerical example $\nu = 5$,

$$q_{0.01}(Y) = \boxed{-2.606464}, \qquad q_{0.05}(Y) = \boxed{-1.560850}.$$

(d) Standardized-t RiskMetrics VaR and comparison

Suppose $R_t = \sigma_t Y_t$ conditionally, with Y_t distributed as the standardized Student's t in part (c). Using the *same* EWMA σ_t from (a),

$$\boxed{\mathrm{VaR}^t_{\alpha,t} = -\sigma_t \sqrt{\frac{\nu-2}{\nu}}\, t_\nu^{-1}(\alpha)}, \qquad \begin{cases} \mathrm{VaR}^t_{5\%,t} = 1.560850\,\sigma_t, \\ \mathrm{VaR}^t_{1\%,t} = 2.606464\,\sigma_t \end{cases} \quad (\nu = 5). \tag{6}$$

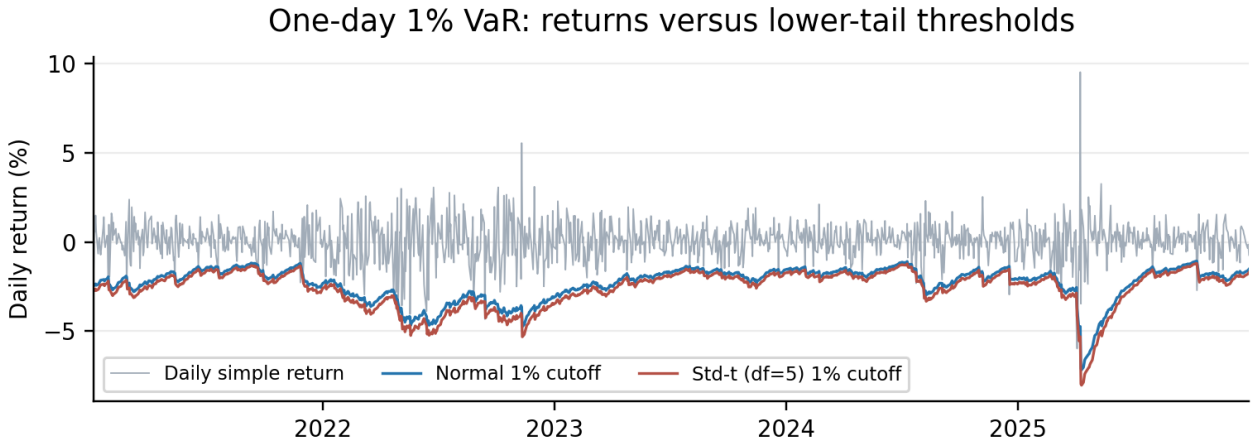


Figure 3: Daily returns and negative 1% VaR thresholds. With $\nu = 5$, the standardized-t model sets a more conservative 1% threshold than the normal model.

VaR comparison and assessment (continuation of part d)

Applying exactly the same exception rule and likelihood-ratio test to all four VaR series gives the following *out-of-sample daily forecasts within the specified window*, subject to the fixed model parameters stated on page 1:

Method	Tail	Expected	Observed	Rate	Kupiec p-value
Normal	5%	62.75	74	5.90%	0.1560
Normal	1%	12.55	26	2.07%	0.0009
Standardized t_5	5%	62.75	81	6.45%	0.0234
Standardized t_5	1%	12.55	19	1.51%	0.0890

Interpretation of the 5% and 1% confidence assessments.

- **5% VaR:** the normal model's observed 5.90% breach frequency is closer to the nominal 5% than the standardized-t_5 model's 6.45%. The normal model passes the 5%-significance Kupiec test ($p = 0.156$), whereas the standardized-t_5 model fails ($p = 0.0234$).
- **1% VaR:** the standardized-t_5 model reduces breaches from 26 to 19 and increases the loss quantile coefficient from 2.326348 to 2.606464. It passes the 5%-significance Kupiec test ($p = 0.089$), while the normal model fails ($p < 0.001$). However, its realized breach rate of 1.51% still exceeds the nominal 1%, so the improvement is incomplete.
- **Why this pattern occurs:** after variance standardization, Student's t with $\nu = 5$ has a *smaller* 5% loss coefficient but a *larger* 1% loss coefficient than the Gaussian model. Heavy tails mainly change the more extreme quantiles. A larger daily positive return also raises the next day's volatility forecast because the RiskMetrics update uses squared returns.
- **Limitations:** unconditional exception frequency alone does not prove that exceptions are independent. A complete risk-model evaluation should additionally consider exception clustering (conditional coverage), model-parameter sensitivity, and economic loss from large breaches. The chosen $\nu = 5$ is an explicit illustration, not an estimated or instructor-given parameter.

Final conclusion. For the 2021–2025 S&P 500 sample and the stated assumptions, *normal EWMA VaR is better calibrated at the 5% level, while standardized-t_5 EWMA VaR is better calibrated at the 1% level*. Neither specification should be described as uniformly best at both confidence levels.

Data sources and clickable companion references.

1. S&P Dow Jones Indices LLC, *S&P 500 [SP500]*, Federal Reserve Bank of St. Louis (FRED): official series and daily source table.
2. Historical price mirror: FRED price CSV mirror. Final 2025 observations were checked against FRED.
3. Reproducible project files: daily VaR calculations (CSV), S&P 500 closing prices (CSV), LaTeX source, and repository documentation.
4. The separately supplied editable `ISOM3540_21089310_ass3_Excel.xlsx` has the `Assumptions`, `Warmup_2020`, `Daily_Data`, `Backtest` and `Excel_Charts` sheets. Its GitHub publication status is recorded in the README.